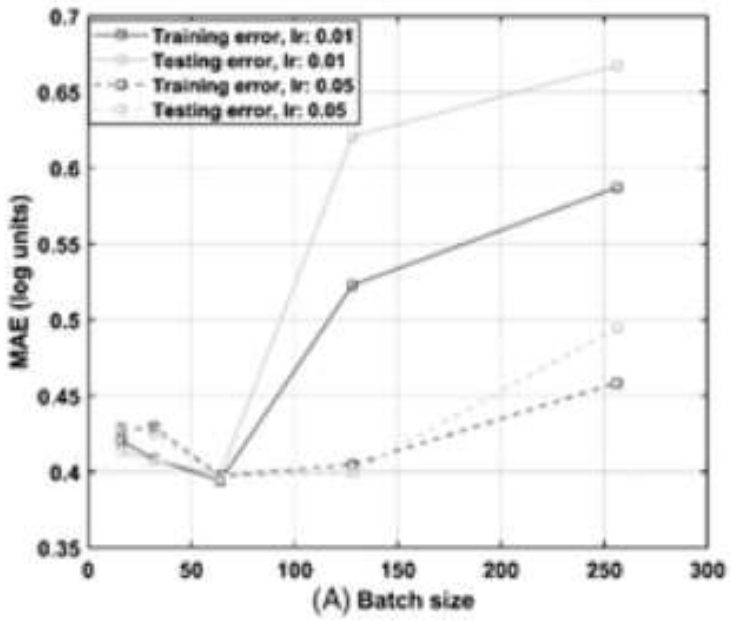
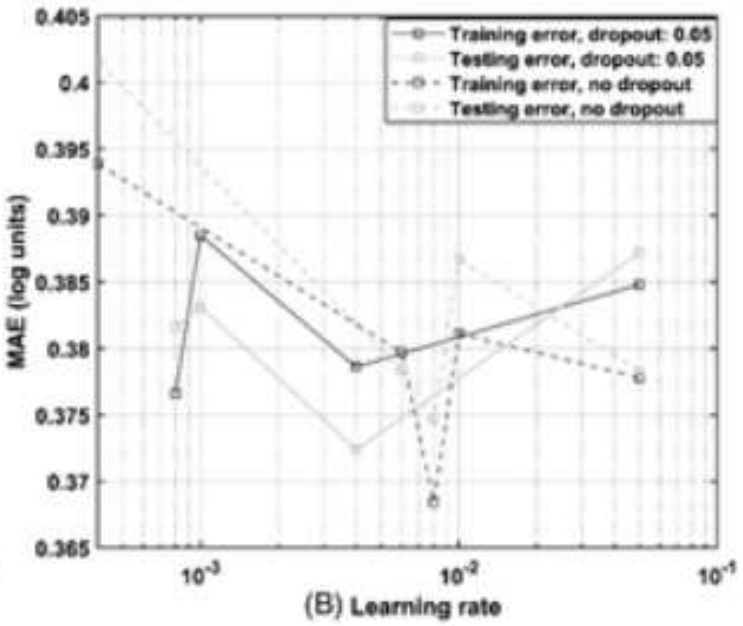
Ground motion prediction equations using ANN

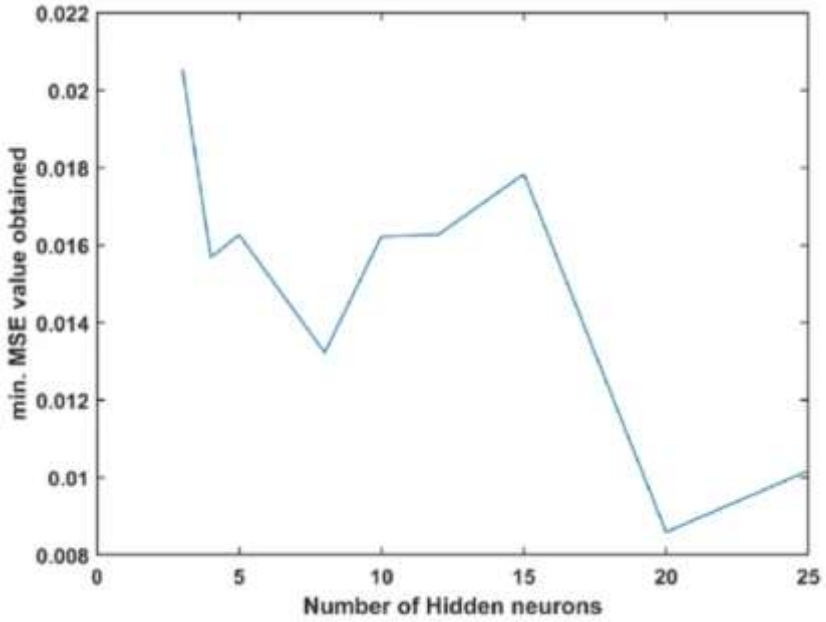
$$\begin{pmatrix} \log_{10}(\mathrm{PGA}) \\ \log_{10}(\mathrm{PGV}) \\ \log_{10}(\mathrm{Sa}_{0.01\mathrm{s}}) \\ \vdots \\ \log_{10}(\mathrm{Sa}_{4\mathrm{s}}) \end{pmatrix} = f(\, M_{\mathrm{w}} \quad \log_{10}(V_{\mathrm{s30}}) \quad R_{\mathrm{rup}} \quad \log_{10}(R_{\mathrm{rup}}) \quad F\,).$$

Use filtered data
Number of Input, output, and hidden neurons
Number of weights and bias

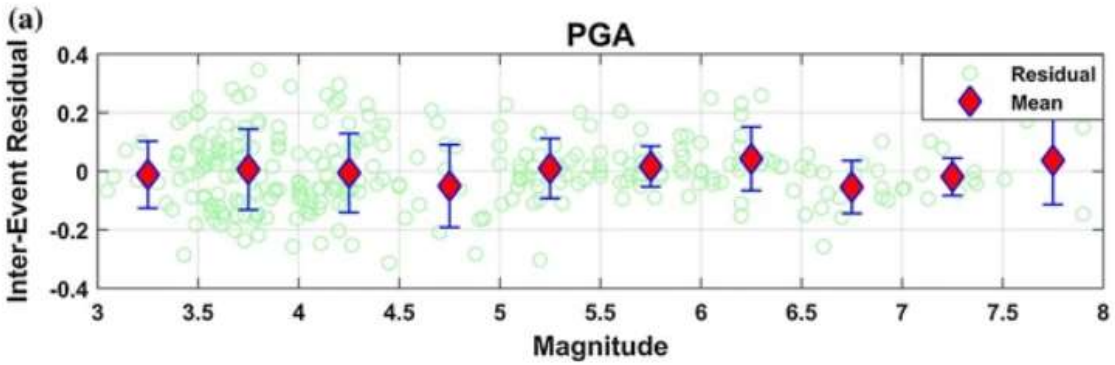
Network Hyperparameters

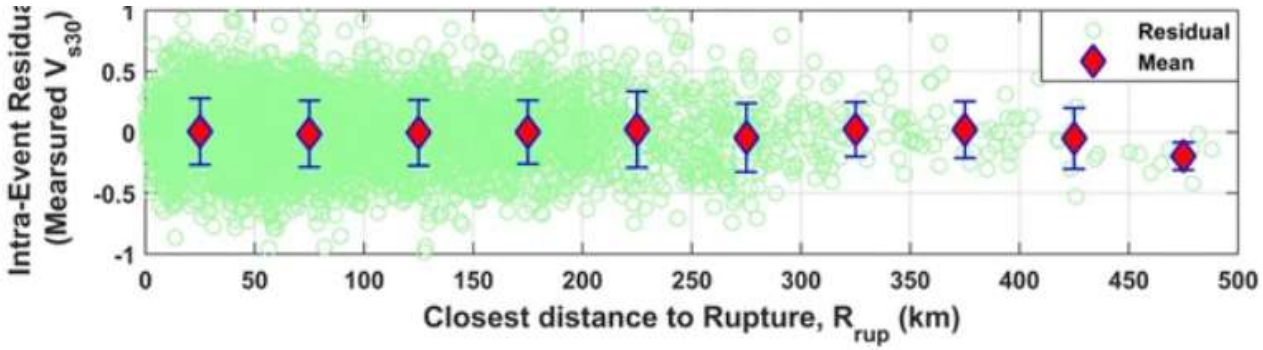


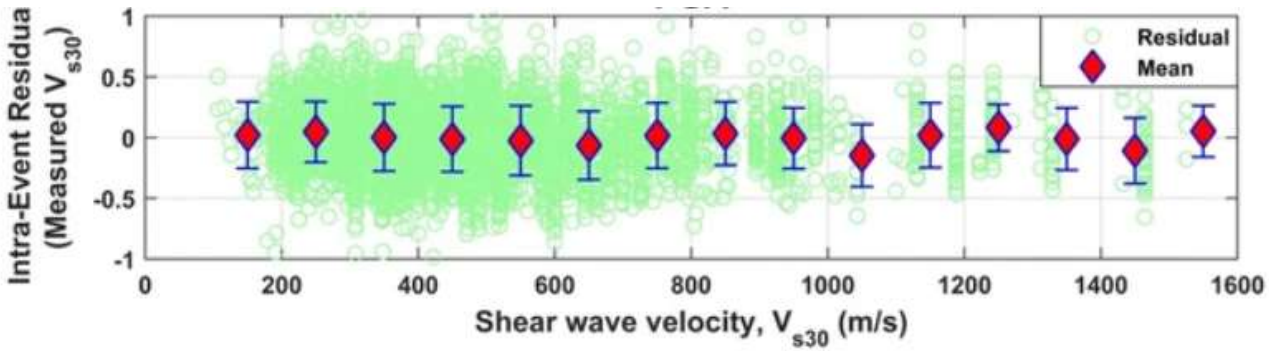




Residual Analysis

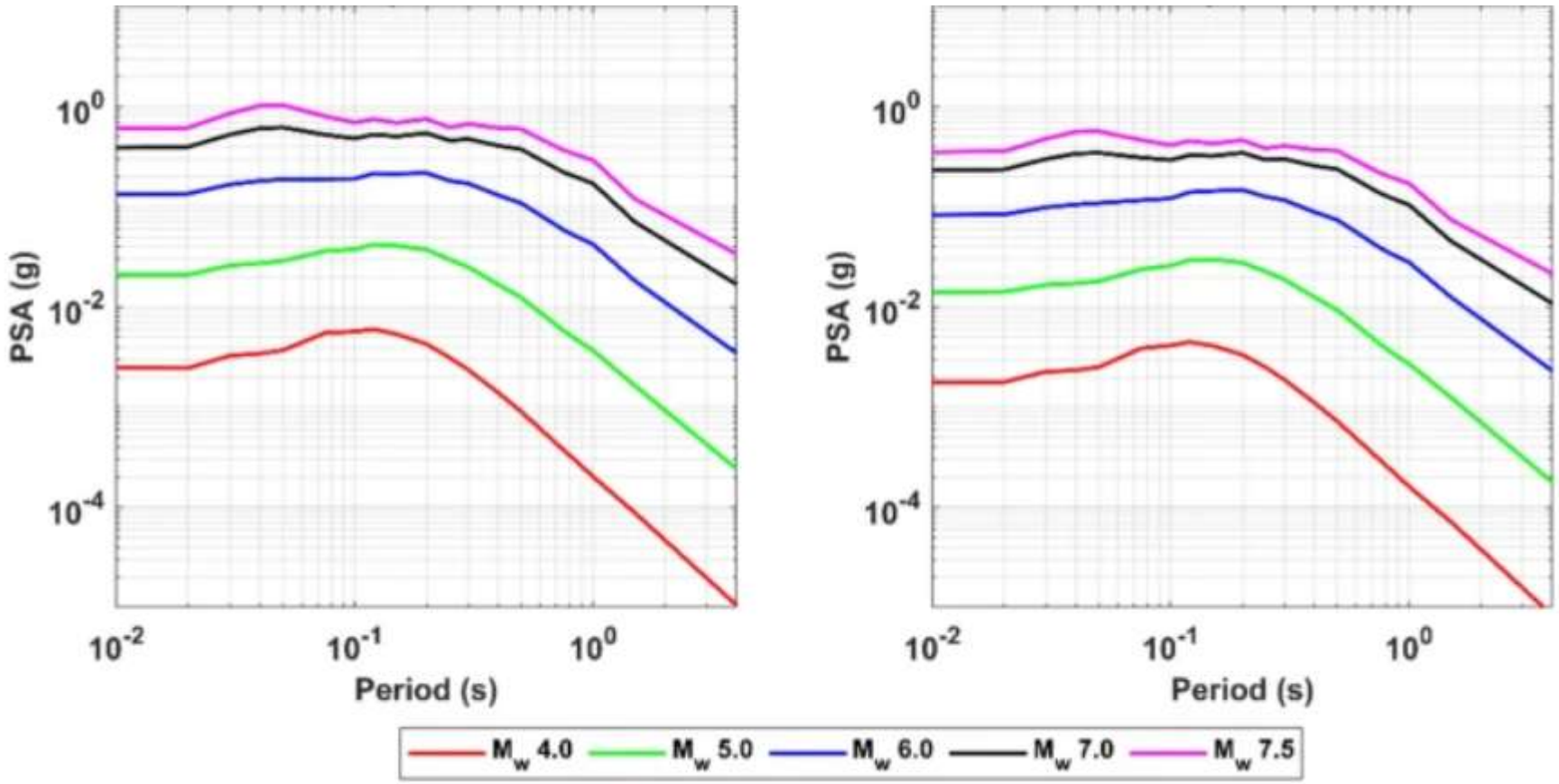




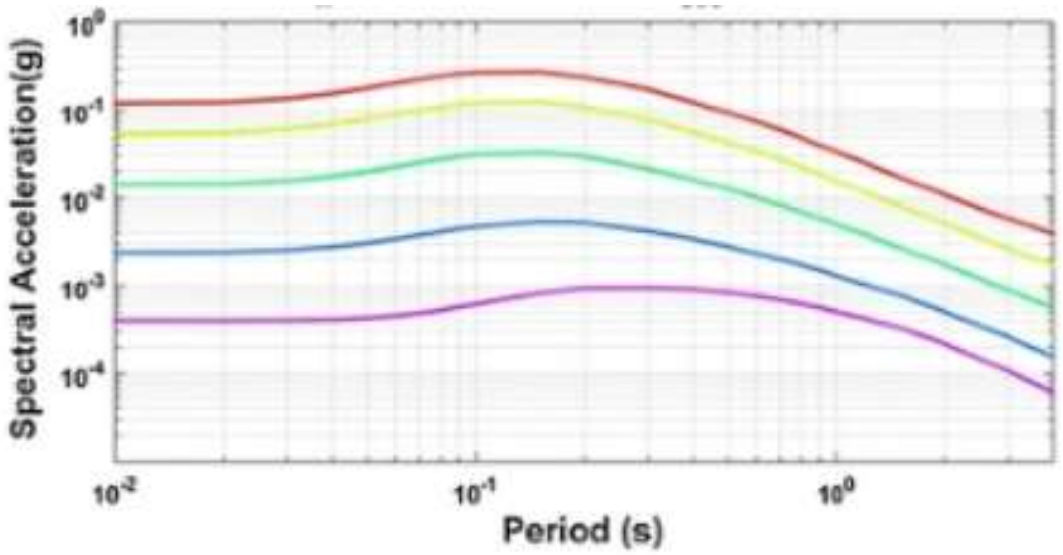


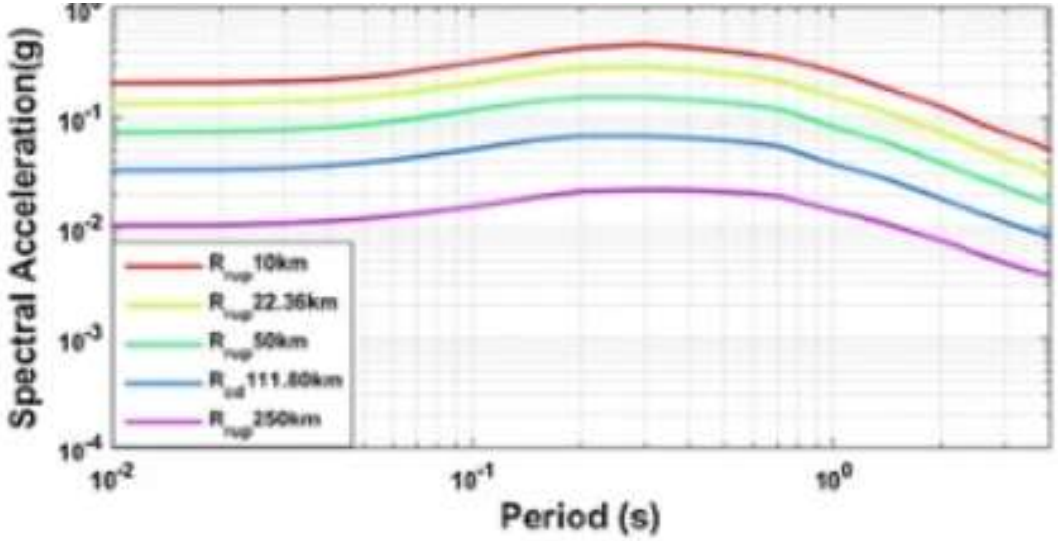
Parametric study

Check w.r.t Magnitude

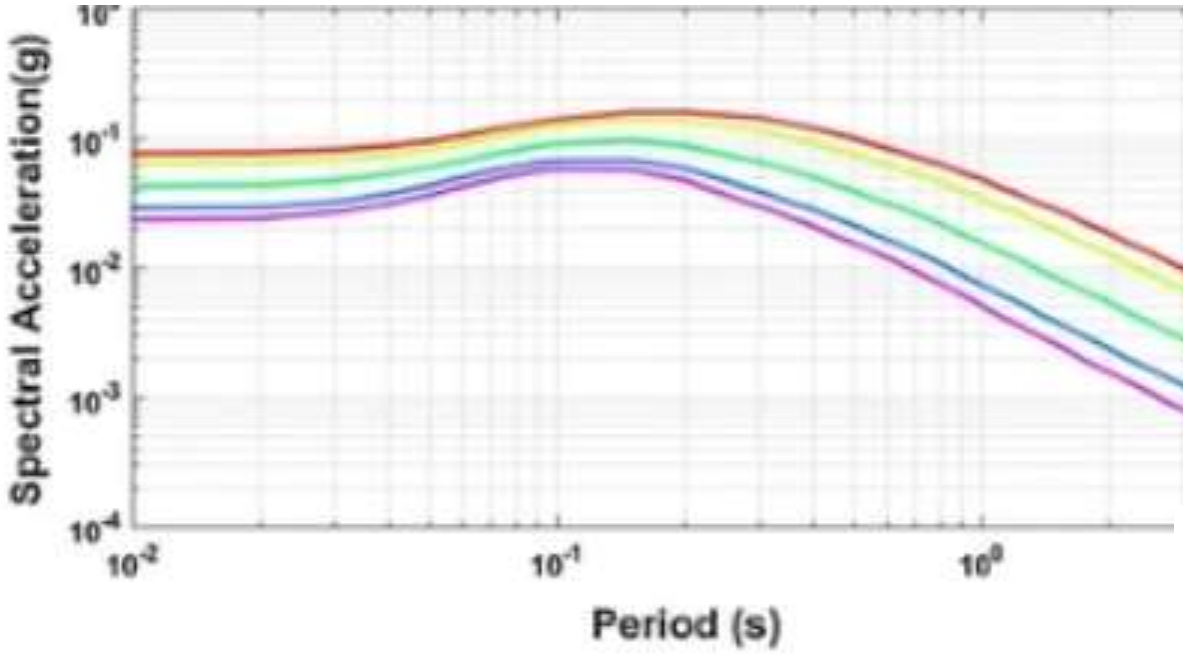


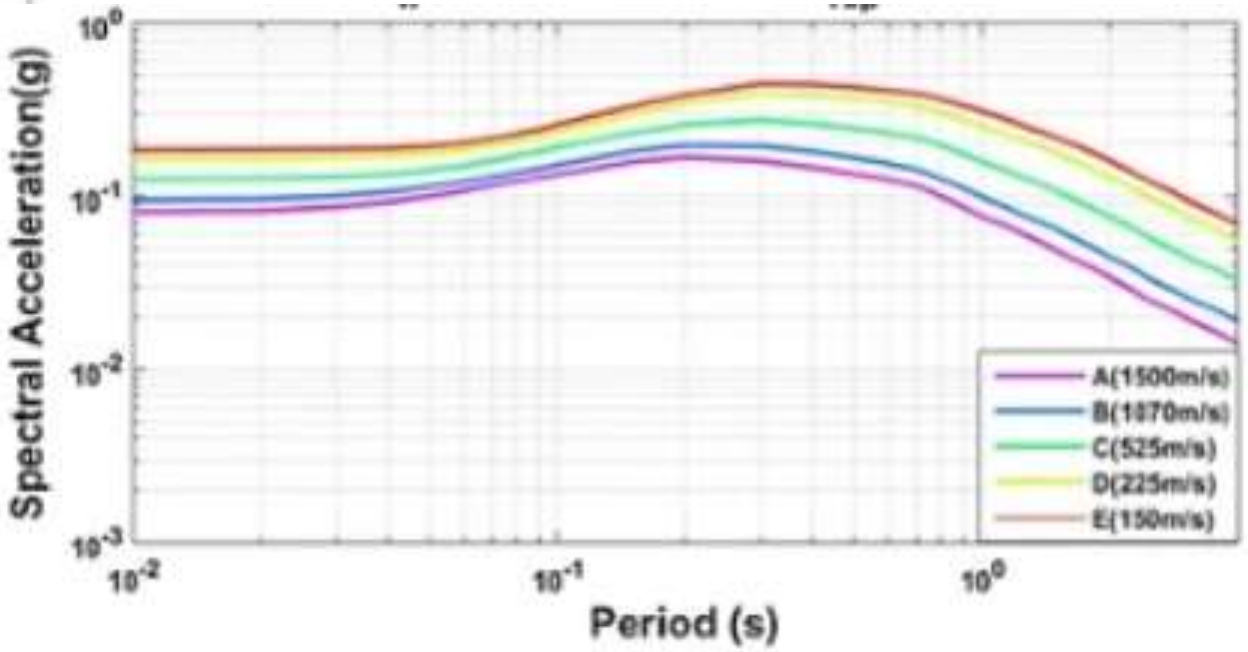
Check w.r.t Rupture distance



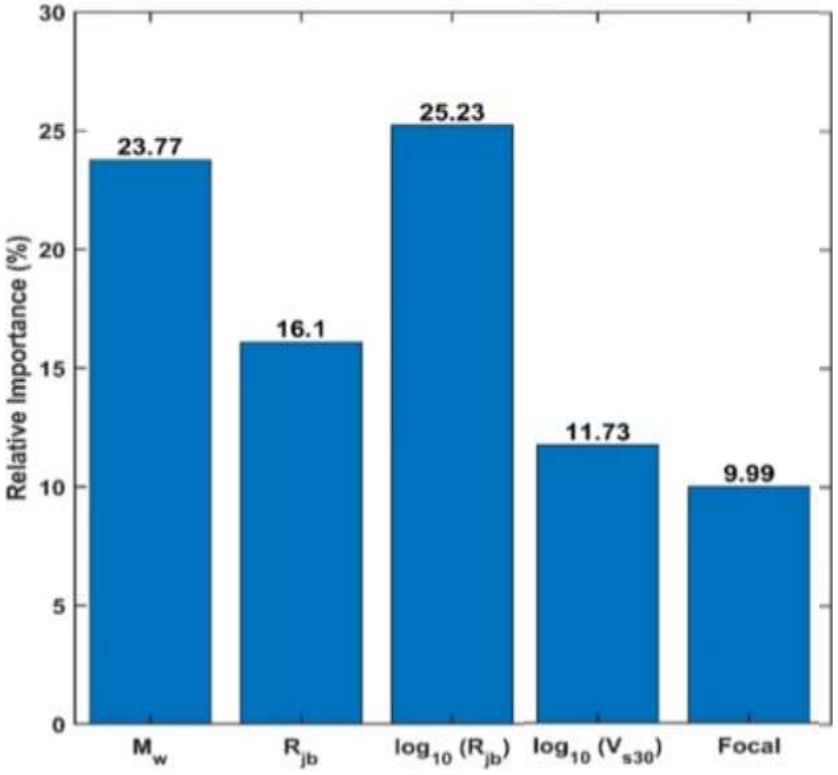


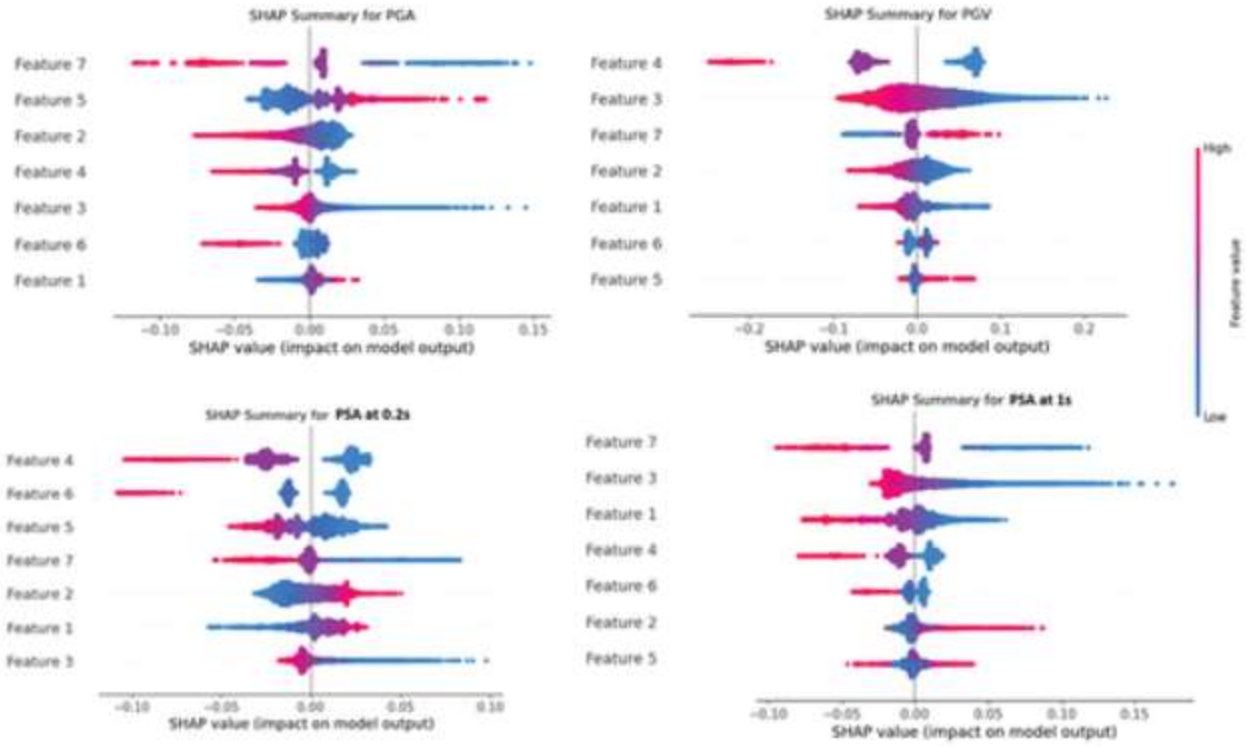
Check w.r.t Velocity





Relative Importance and SHAP Analysis





Todays Tasks:

1. Complete the ANN coding and generate all the graphs as mentioned.
2. Work on additive neural networks and plot the required graphs.
3. Write your insights based on the generated plots.
4. Divide yourselves into 10 groups and update the details in the Excel sheet.